Poisson regression

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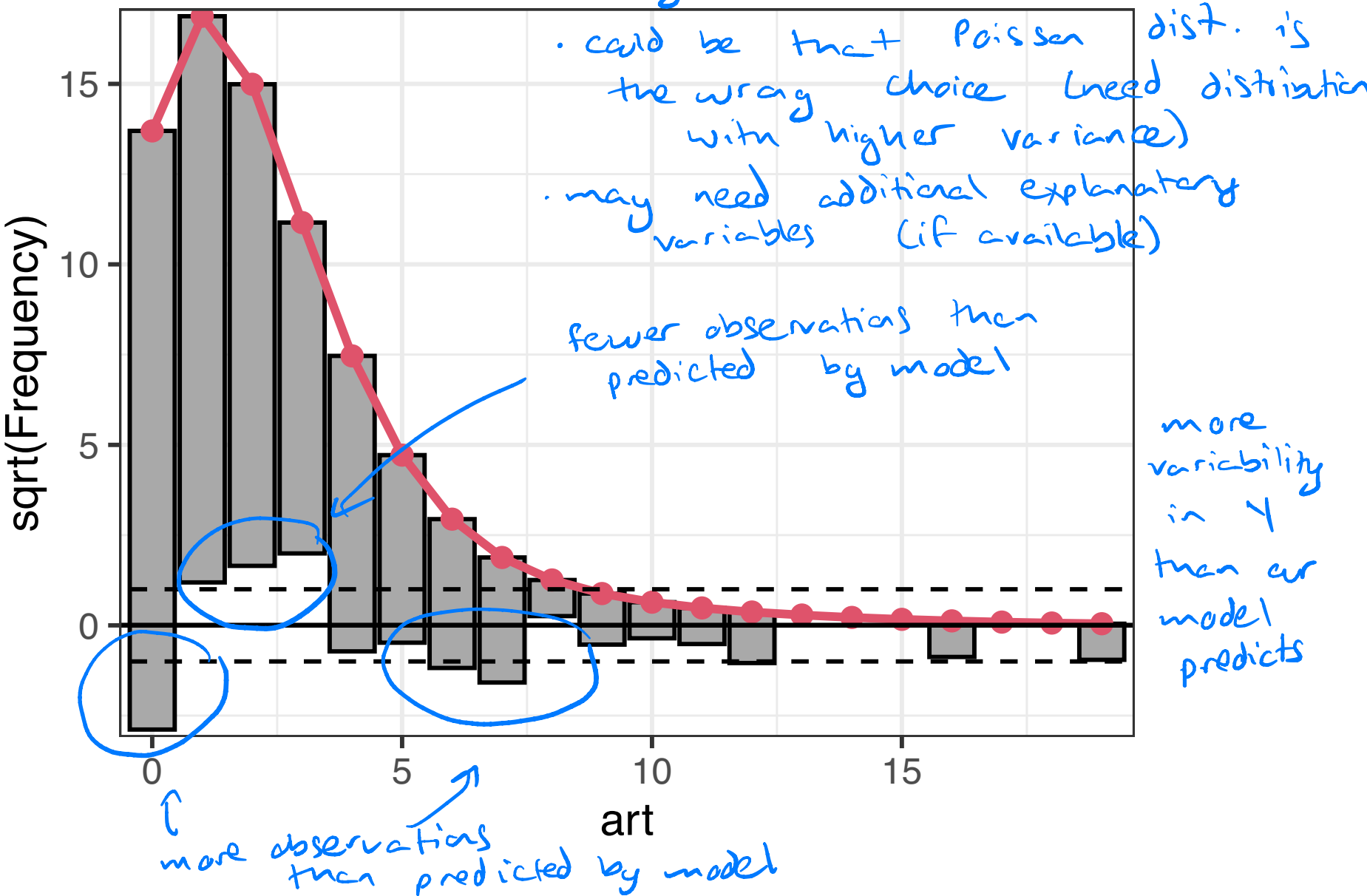
Activity

Use the provided packet to answer the following questions:

- Does the Poisson regression model appear to be a good fit?
- If not, which piece of the model do you think is mis-specified?

Rootograms: an overall assessment of model fit

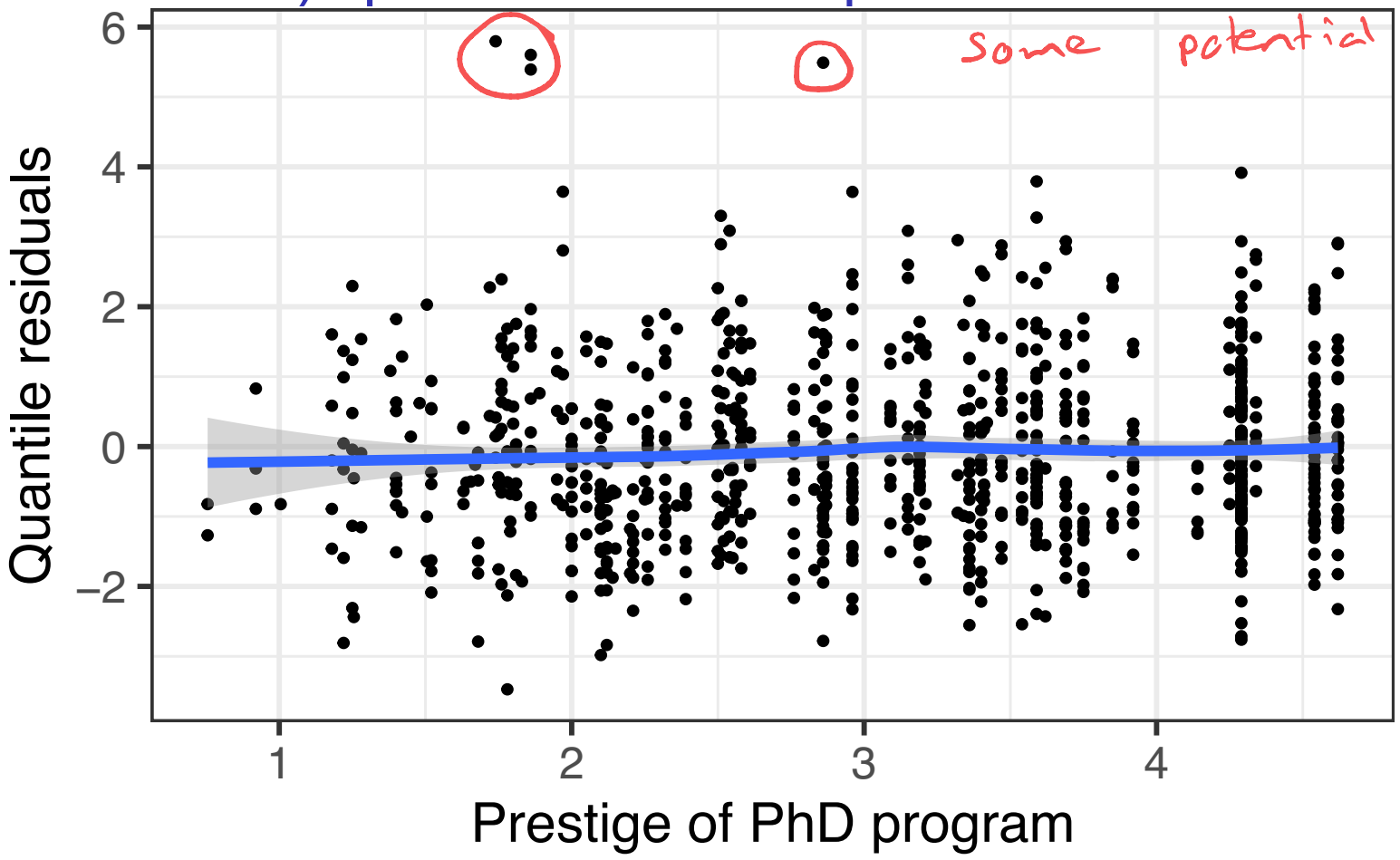
```
topmodels::rootogram(m_art)
```



Rootograms: an overall assessment of model fit

- horizontal axis: possible values for the response variable
- vertical axis (square root scale): predicted count for the model (points) and *observed* counts from the data (bars)
- Assessing whether predicted counts match observed counts
- Lack of fit could be due to:
  - Wrong choice of distribution
  - Additional explanatory variables need to be added to model
  - Issues with shape assumption

(Randomized) quantile residual plots



What I am looking for in a quantile residual plot?

- random scatter around 0
- ≈ constant variance
- most quantile residuals between −2 & 2

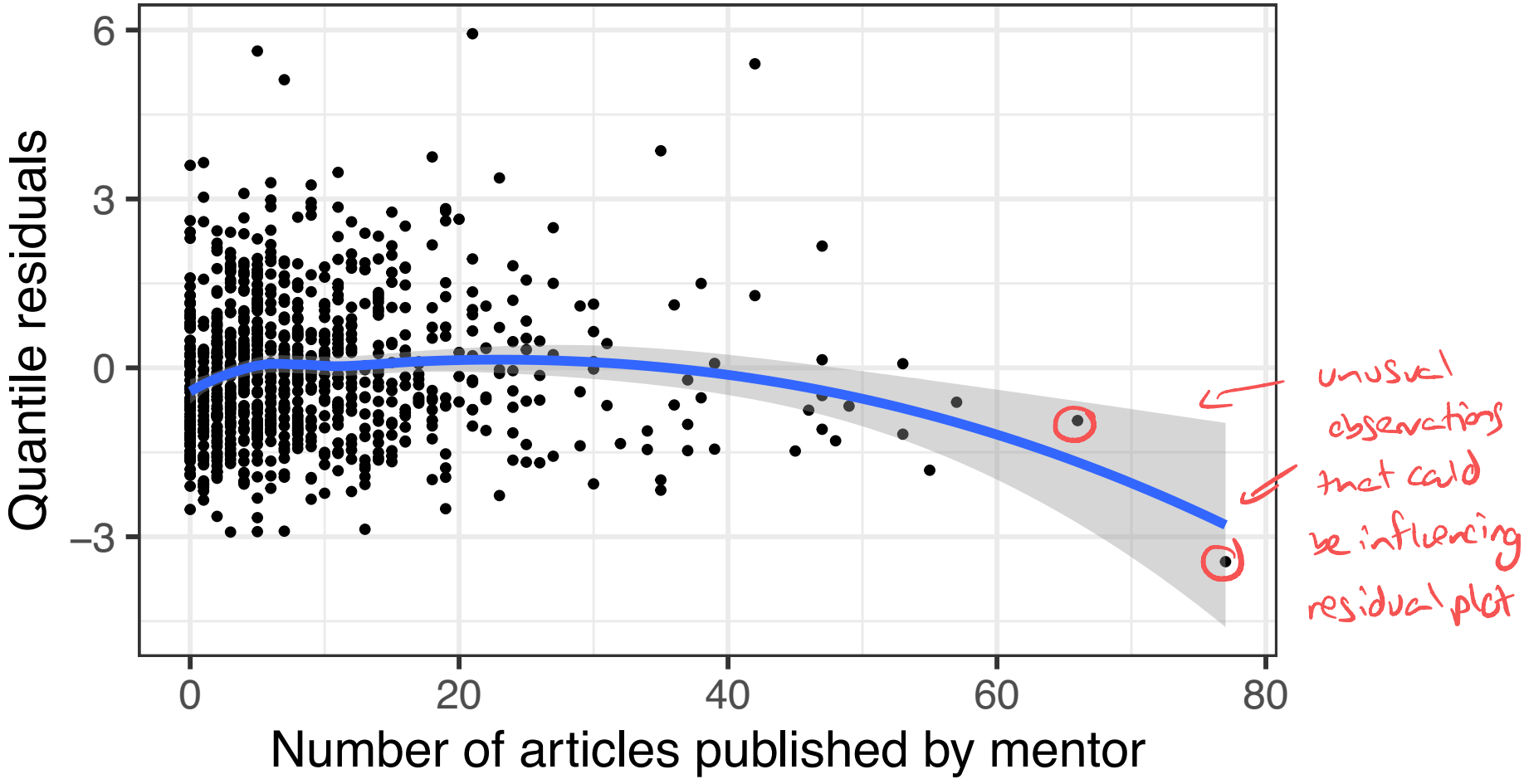
Examining influential points

Cook's distance: How much do my fitted coefficients change if I remove i^{th} observation?

	art	phd	ment	kid5	cooks_distance
328	1	1.78	77	1	0.5587585
915	19	1.86	42	0	0.4113048
914	16	1.74	21	0	0.1478051
911	11	2.86	7	2	0.0905633
912	12	4.29	35	1	0.0638557
913	12	1.86	5	1	0.0625560
909	9	1.86	47	1	0.0481382
669	2	4.54	55	0	0.0351612
803	4	4.54	66	2	0.0243971
883	6	2.51	11	2	0.0199222

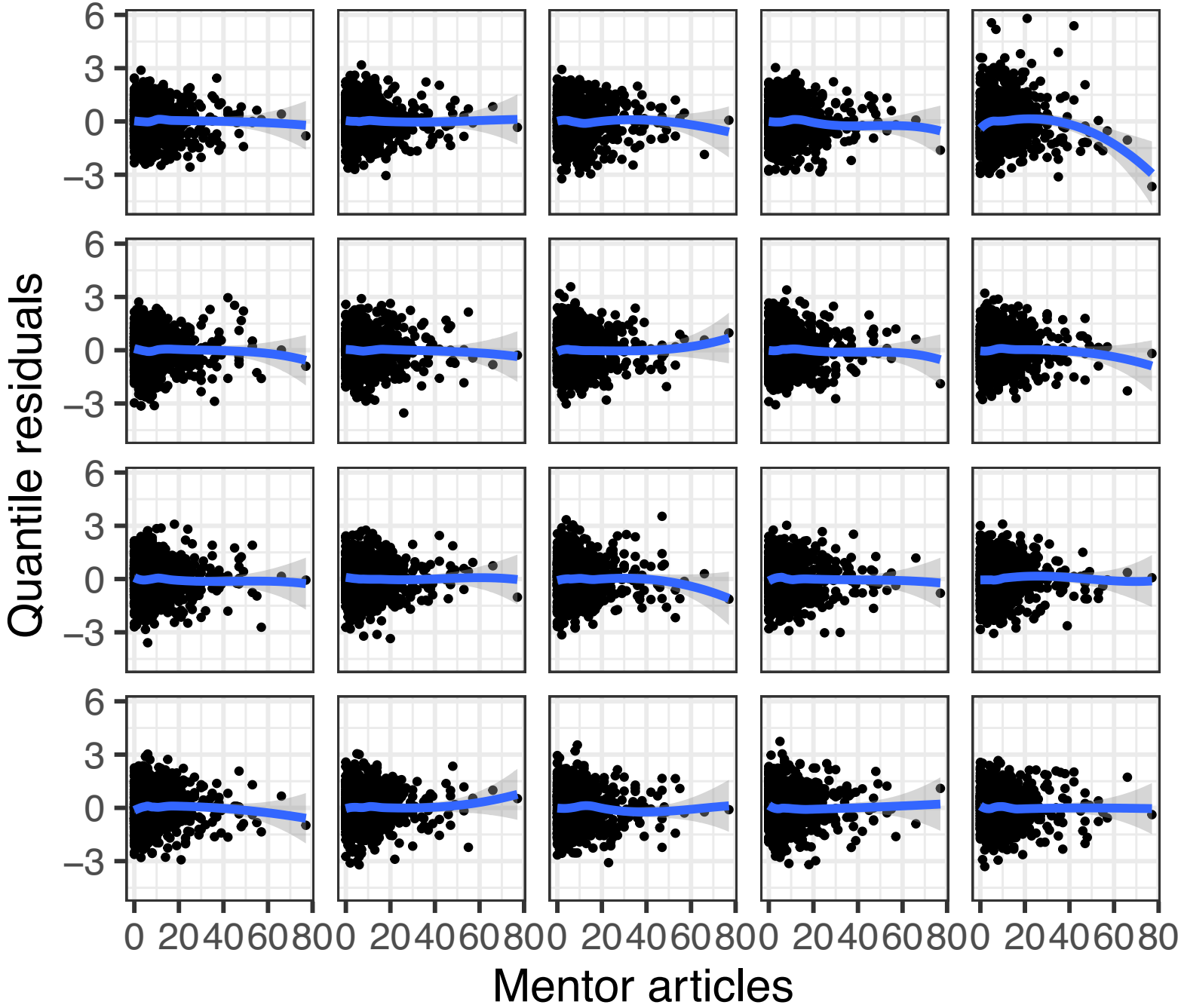
very unusual values, but not clear errors → report results with and without unusual points

(Randomized) quantile residual plots



How could I decide whether deviations in the quantile residual plot are just due to random chance?

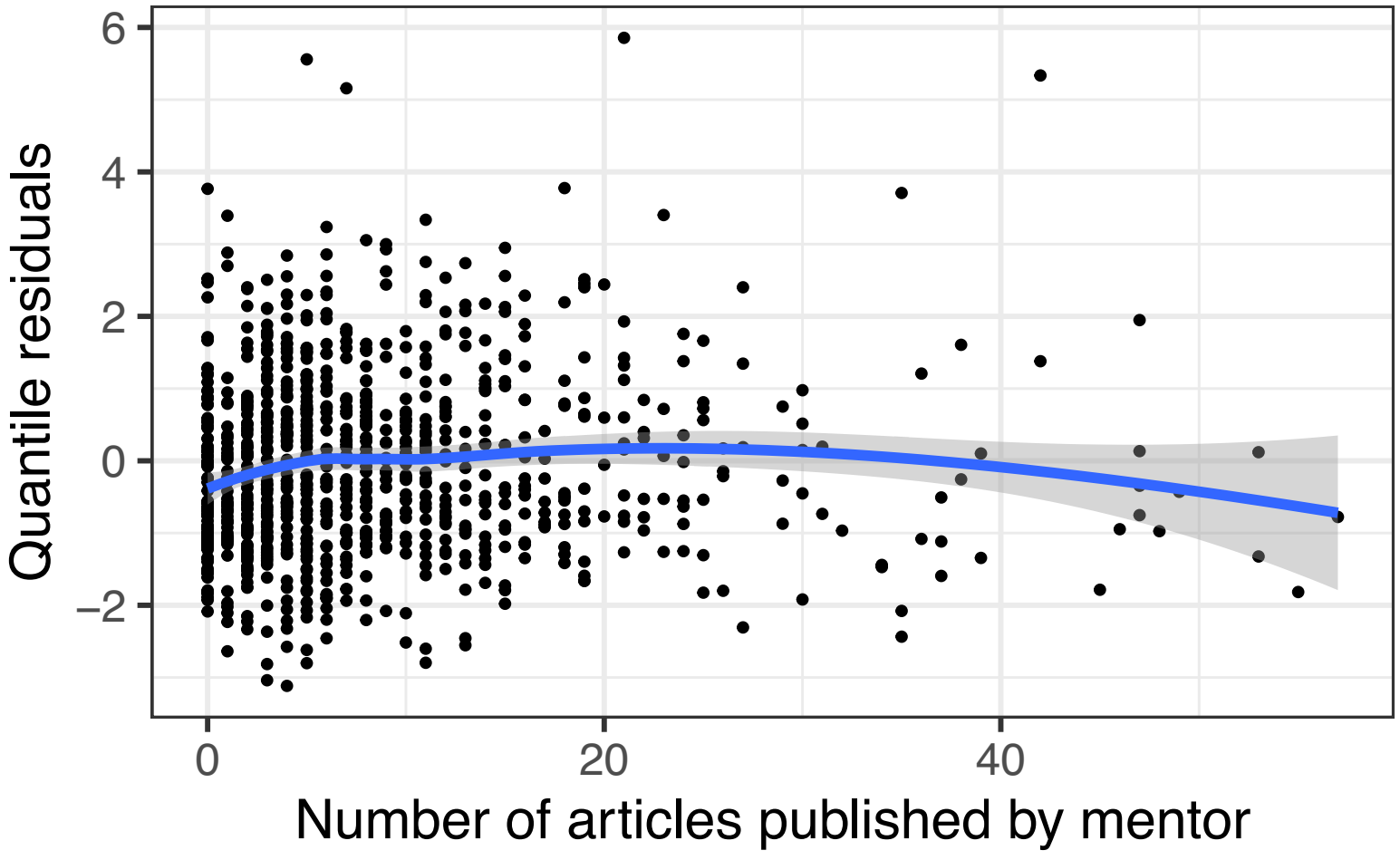
Plot lineup



Quantile residual plot

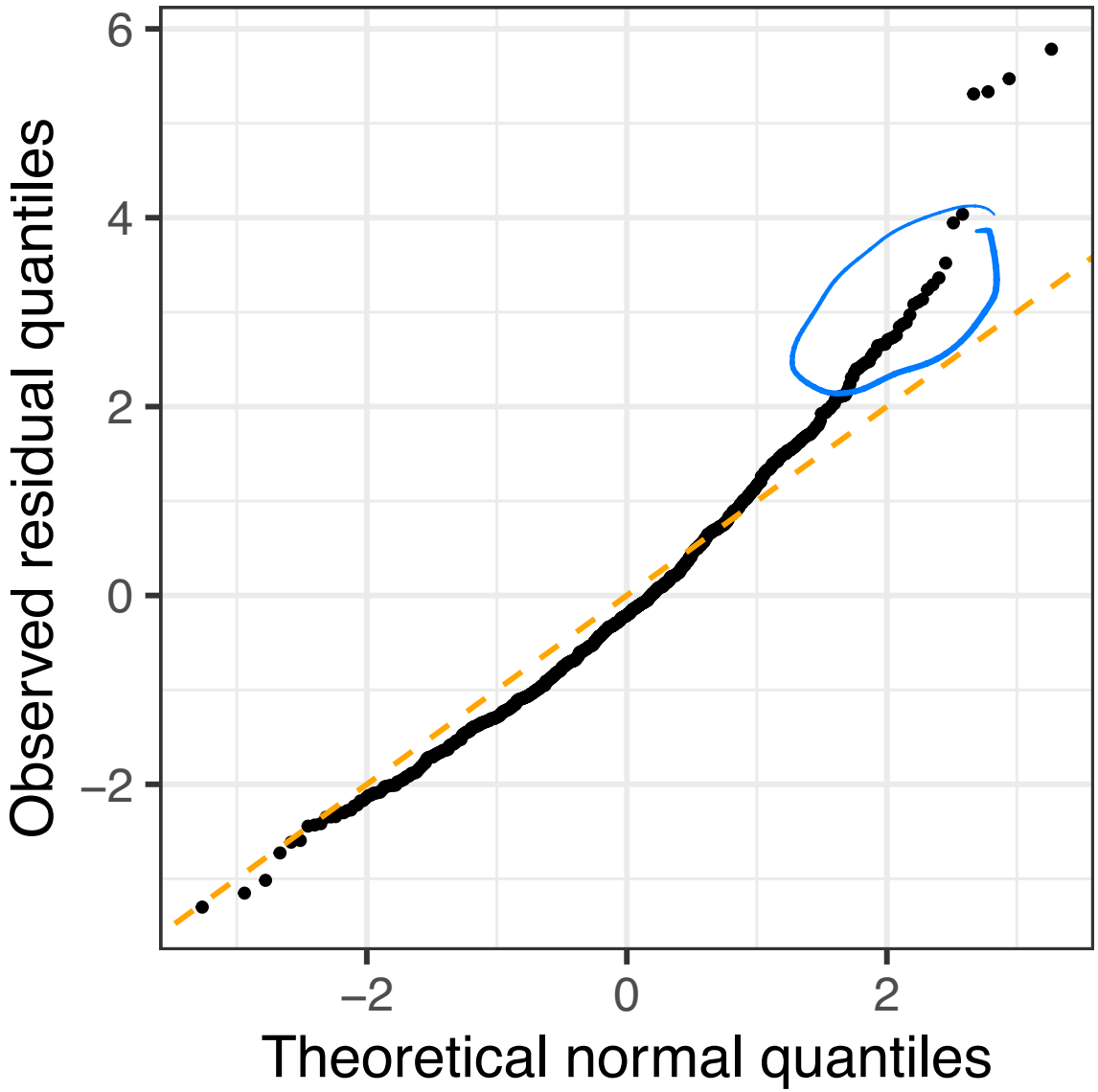
Plot without the two rows with the most number of mentor publications:

Looks better. Shape assumption seems reasonable (other than two outliers)



QQ plot

If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$.



More large, positive residuals than expected

⇒ some right skew, greater variability in residuals than expected

(matches rootogram)

A goodness-of-fit test

H_0: model is a good fit to the data

H_A: model is not a good fit to the data

Idea: find a test statistic that describes overall model fit

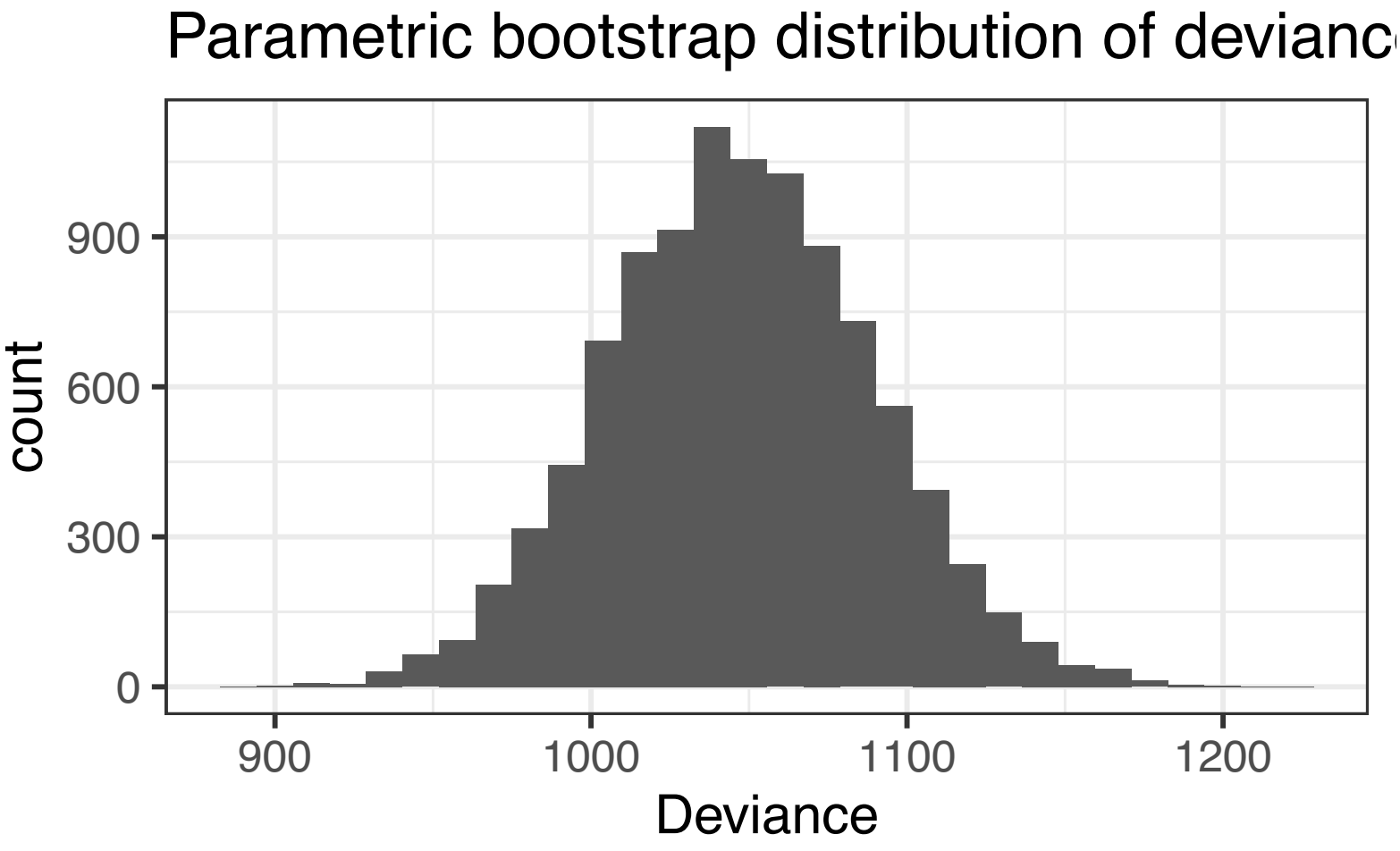
teststat: (residual) deviance

Null distribution??

Parametric bootstrapping

- Fit model (full model)
  $\rightarrow \hat{\lambda}_1, \ldots, \hat{\lambda}_n$
- Simulate new Y_i^* from fitted model
  $Y_i^* \sim \text{Poisson}(\hat{\lambda}_i)$
  (so H_0 must be true for simulated data)
- Re-fit model on $(x_1, Y_1^*), \ldots, (x_n, Y_n^*)$
- Calculate test statistic (deviance) on model with simulated data
- Repeat!

Parametric bootstrapping




```
m_art$deviance
```

```
## [1] 1661.067
```

very unusual if H_0 is true!

Approximate null distribution

If the model is a good fit to the data, then the residual deviance approximately follows a χ^2 distribution, with the same degrees of freedom as the residual deviance.

```
m_art$deviance
```

```
## [1] 1661.067
```

```
m_art$df.residual
```

← n - p

```
## [1] 911
```

if deviance >> n - p, suggests model is not a good fit

(mean of χ^2_d = d, variance = 2d)

Summary so far

For the Poisson regression on the articles data, here is my perspective:

- There are several potential outliers
- The rootogram shows our predicted counts are somewhat low in the tails of the distribution, compared to the observed counts
- Quantile residual plots suggest the shape assumption is overall pretty reasonable
- The QQ plot suggests that variability of the residuals is higher than expected
- The goodness-of-fit test suggests that some part of our model is mis-specified

Question: If this particular model is mis-specified, what do you think is the cause? How could we modify the model?

want a different distribution for Y (e.g., some modification of Poisson with greater variance) → quasi-Poisson, Negative Binomial